# Does Morphology-Aware Tokenization Improve Explainability? A Russian Sentiment Analysis Case Study

**Alexa Garcia Nunez**
a.garcianunez@se25.qmul.ac.uk
QMUL Student ID: 260139377

## ABSTRACT

Standard Byte-Pair Encoding (BPE) is a frequency based tokenizer rather than linguistically informed one. In Russian, a morphologically rich language, this can split words which carry sentiment into arbitrary subword fragments. Since token-level explainability methods such as SHAP (SHapley Additive exPlanations) explain model predictions in terms of the tokens given to the classifier, these tokens boundaries may affect how linguistically interpretable an explanation is. This project investigated whether morphology-aware tokenisation could produce more linguistically interpretable explanations for Russian sentiment classification while maintaining broadly comparable predictive performance.

This was explored using the RuSentiment social-media dataset and two matched lightweight RoBERTa classifiers. The control pipeline used standard SentencePiece BPE, while the morphology-aware pipeline used a semi-supervised Morfessor model with BPE fallback for unseen or unsupported segments. The final morphological model achieved a held-out boundary F1 of 0.874 and an exact segmentation match of 0.852. Standard BPE achieved a test macro F1 of 0.7353, compared with 0.7243 for the morphology-aware model, representing a small reduction in predictive performance.

The SHAP analysis showed that morphology-aware tokenisation can improve linguistic readability when it preserves an important sentiment-bearing unit. However, this improvement was not universal. In the cases where standard BPE already produced meaningful tokens there was little benefit, while incorrect morphological segmentation could make explanations less coherent. The findings therefore suggest that morphology-aware tokenisation can improve explainability, but that the benefit depends strongly on segmentation quality, an issue which persists within the field computational linguistics.

## 1. INTRODUCTION

Neural NLP classifiers normally use tokens rather than complete words. One of the most common subword tokenisation methods is Byte-Pair Encoding (BPE), a statistical subword method adapted to neural NLP by Sennrich et al. [1]. BPE creates a fixed vocabulary by repeatedly combining character sequences that frequently occur together. This is useful because a model can represent frequent words efficiently while also constructing rarer or previously unseen words from smaller known subword units, rather than requiring every possible word to have its own vocabulary entry. However, the boundaries created by BPE are based on statistical frequency rather than the linguistic structure of a word [1].

This distinction is particularly relevant for Russian because it is a morphologically rich language in which stems, prefixes and suffixes contribute to grammatical and lexical meaning. Russian often preserves visible relationships between words through recurring roots and affixes. For example, ужас means “horror” is also a common root forming the base of many other words. For example , ужасный (“terrible”), ужасно (“terribly”) and ужасаться (“to be horrified”) all contain the recurring root ужас-, even though their natural English translations use different lexical forms.

This makes Russian a useful setting for testing whether morphology-aware tokenisation can preserve linguistically meaningful structure that a purely frequency-based tokenizer may split arbitrarily. Using the same example, ужасный (“terrible”) may be represented by standard BPE as ужа | с | ный. Although a classifier may successfully learn from these tokens, ужа and с are not particularly useful linguistic explanations for a Russian speaker reading the output. A more recognisable division is ужас | ный, where ужас (“horror”) is itself a meaningful lexical root as mentioned.

This becomes important when post-hoc explainability methods are applied. SHAP, uses shapley values to assi. importance to the features supplied to a model. In an NLP context, these features are the token positions. Therefore, if an important sentiment word is divided into arbitrary statistical pieces, SHAP can produce a numerically valid attribution that is nevertheless difficult to interpret linguistically. It is important to emphasize that this project is not asking whether morphology-enriched tokenizing approaches produce larger SHAP values. Instead, it asks whether the tokens that do receive higher attribution are more meaningful to a human reader.

Hence, this projects main research question is as follows:

> **Does morphology-aware tokenisation improve the linguistic interpretability of explanations for Russian sentiment classification while maintaining comparable predictive performance?**

To investigate this, two similar sentiment-classification pipelines were developed. Both use the same RuSentiment data, the same lightweight RoBERTa architecture and the same training procedure. The main experimental difference is the tokenisation strategy. The control uses standard SentencePiece BPE, while the second introduces morphological segmentation before a modified BPE stage. Classification performance is not the main focus of the project. Instead, it is used to check that introducing morphology does not substantially reduce sentiment-classification performance, so that the two models remain suitable for a fair explainability comparison.

RuSentiment is a Russian sentiment classification dataset and is useful because it consists of real Russian social-media posts rather than clean dictionary text [6]. The data contains informal spelling, slang, punctuation, emoticons and other expressive features typical of online communication. These properties make morphological segmentation harder, but also make the experiment more realistic. The initial project proposal considered both SHAP and LIME as XAI ( explainable AI) methods worth exploring, however the final implementation focused on only SHAP so that a deeper controlled analysis of token-level linguistic attribution could be completed within the project scope.

The main challenges in this project came from the fact that morphology-aware tokenisation is not automatically linguistically correct. An unsupervised segmenter can identify recurring word parts that are statistically useful but not genuine morphemes, and noisy social-media spelling makes this harder. A second challenge is that morphological annotations do not always map directly onto the exact written form of a word, which complicates their use for tokenizer training. A final challenge was how to preserve useful Morfessor segments without losing the ability to represent rare or previously unseen social-media forms. This motivated the final hybrid design, in which Morfessor-derived segments observed during training are retained as whole tokens, while BPE provides fallback coverage when a new segment cannot be represented directly.

## 2. Related Work

Previous work on morphology-aware NLP has explored statistical segmenters, supervised neural systems and modifications to standard subword tokenisers. Methods such as BPE-knockout remove merges that conflict with morphological boundaries [3], while MorphBPE constrains BPE so that merges respect supplied morpheme boundaries and evaluates the approach on morphologically rich languages including Russian [4]. These studies show that linguistic

information can be introduced into subword tokenisation without completely abandoning the practical advantages of BPE.

Morphological segmentation itself has no single standard solution. The SIGMORPHON 2022 Morpheme Segmentation shared task compared a range of systems across 9 languages, including Russian, from general-purpose approaches such as segmentation algorithms to specialised supervised models [5]. The task showed that the highest performing segmenters were complex neural networs. This approach could therefore have been used here, but this would introduce another substantial learned model into an experiment whose main purpose is explainability rather than developing a complex Russian morphological analyser.

'Morfessor' was hence chosen as a compromise between ordinary frequency-based tokenisation and a fully supervised morphological model. Morfessor is a statistical morphological segmentation toolkit developed by Creutz and Lagus [2] It learns a reusable inventory of recurring word parts from the vocabulary supplied to it. For example, if the corpus contains ужас, ужасный, ужасно and related forms, repeatedly representing these words using the component ужас can give a more economical description of the vocabulary. Morfessor does not know that ужас is a Russian root; instead, recurring linguistic structure creates recurring statistical patterns from which actual morpheme units are more likely to emerge [2]. However it's unsupervised predictions are still not guaranteed to be linguistically correct all the time. This limitation motivated the later addition of labelled SIGMORPHON supervision.

SIGMORPHON is a research community focused on computational morphology, and its 2022 shared task released gold-standard word segmentations for nine languages, including Russian. Here, "gold standard" means the reference segmentation against which a system prediction is evaluated. For example, if the reference segmentation has a boundary after ужас, a Morfessor prediction is correct at that position when it places the same boundary there. These labels are morphological reference labels and are separate from the RuSentiment sentiment labels.

Explainability research provides the other side of the problem. SHAP, introduced by Lundberg and Lee [7], assigns contribution values to the features used by a model. In NLP, SHAP is commonly applied at token level, meaning that the tokens produced by the tokenizer become the features receiving attribution [7], [8]. This makes tokenisation important to explanation quality because changing the token boundaries also changes the units through which the model's behaviour is explained.

Recent work by Yagiz and Horasan provides particularly relevant evidence that tokenisation can directly affect explainability in morphologically rich languages. Their MAFEX framework identifies what they call 'Tokenization-Morphology Misalignment', where standard subword tokens do not correspond well to meaningful morphemes and token-level attribution becomes dispersed or difficult to interpret [9]. In their Turkish language experiments, including sentiment classification, morpheme-aligned explanations substantially outperformed standard token-level baselines such as SHAP, Integrated Gradients and DeepLIFT at identifying the key linguistic unit responsible for a prediction. Their method, however, keeps the original model and tokenizer and instead projects the explanation onto morpheme-aligned units after attribution has already been calculated. This is different from the present project, which changes the tokenizer itself before model training and then compares the resulting SHAP explanations directly. Their findings therefore support the broader hypothesis that morphology matters for XAI, while this project contributes a complementary Russian sentiment case study that tests whether improving the representation at tokenisation time can make commonly used post-hoc XAI methods such as SHAP easier to interpret.

The interaction between tokenisation, morphology and XAI is a complex, ambitious and relatively under-researched area. Hence, this project does not aim to provide a complete solution but instead contributes a small but important controlled Russian sentiment case study to an emerging research gap. By keeping the classifier architecture fixed and changing the tokenisation strategy, it provides a focused way of examining whether better linguistic structure at the input level can make SHAP explanations easier to interpret.

## 3. Proposed Methodology

This project uses a controlled comparison between two Russian sentiment-classification pipelines that differ in how they tokenize the input text. The first pipeline uses standard SentencePiece Byte-Pair Encoding (BPE), while the second introduces morphological information before and during BPE tokenisation. Both token sequences are

then passed to matched lightweight RoBERTa classifiers trained from scratch. Keeping the classifier architecture and training procedure constant allows the effect of the tokenisation strategy to be examined separately from changes in model capacity.

The two experimental pipelines can be represented as:

Standard: $x \rightarrow N(x) \rightarrow BPE \rightarrow C$

Morphology-aware: $x \rightarrow N(x) \rightarrow M(x) \rightarrow BPEfallback \rightarrow C$

Where $N$ is shared normalisation, $M$ is the semi-supervised Morfessor segmentation model, $BPE$ is the standard SentencePiece BPE tokenizer and $BPEfallback$ represents the morphology-aware tokenizer in which Morfessor segments seen in training are kept whole and BPE is used only for segments that cannot be represented directly. C is the RoBERTa sentiment classifier.

## 3.1 Data and Preprocessing

The downstream task uses the RuSentiment dataset, which contains Russian social-media posts labelled for sentiment. The final experiment keeps four classes: negative, neutral, positive and speech. After cleaning and removing duplicates, the data was divided into 20,440 training examples, 3,608 validation examples and 2,621 held-out test examples. The same data split is used for both tokenisation conditions so that the comparison remains controlled.

The sentiment label set can therefore be written as: Y = {negative, neutral, positive, speech}

For an input post , the classifier predicts the class with the highest probability: $\hat{y} = \text{argmax}\ P(y = c \mid x), \quad c \in Y$

Before classification, the tokenizer converts the normalised post into a sequence of tokens: $T(x) = (t_1, t_2, \ldots, t_n)$

The main experimental manipulation is therefore the tokenisation function . The standard pipeline produces this sequence using SentencePiece BPE, while the morphology-aware pipeline first applies Morfessor and then uses BPE only as fallback for segments that cannot be represented directly. The classifier architecture and training procedure remain matched between the two conditions.

The development data was split 85/15 using stratification and random seed 42. The same normalisation procedure was applied in both conditions: text was lowercased and separated into Cyrillic strings, Latin strings, numbers and punctuation. Punctuation and emoji were retained because they can contain genuine sentiment information in social-media text.

## 3.2 Evaluation design

The quality of morphological segmentation and linguistic representation is evaluated independently from sentiment classification so that it can be checked before interpreting SHAP. If Bp is the set of predicted boundaries and Bg is the gold-standard set, boundary precision measures how many predicted boundaries are correct, recall measures how many gold boundaries were recovered, and F1 balances the two:

$$\text{Precision} = |Bp \cap Bg| / |Bp|$$

$$\text{Recall} = |Bp \cap Bg| / |Bg|$$

$$F1 = 2 \times (Precision \times Recall) / (Precision + Recall)$$

Exact match is a stricter metric, it only counts a word when the entire predicted segmentation matches the reference. For example, if the gold segmentation is ужас | ный, then ужас | ный is an exact match, while ужа | с | ный is not. Sentiment classification is evaluated with accuracy and macro F1. Explanation quality is assessed qualitatively by examining whether high-attribution SHAP features correspond to recognisable and sentiment-relevant linguistic units rather than by comparing which model has numerically larger SHAP values.

## 4. IMPLEMENTATION

The methodology was implemented using Google Colab. The final notebook loads and cleans RuSentiment, trains and evaluates Morfessor, constructs both tokenisers, trains the two matched classifiers, evaluates them and finally generates token-level SHAP explanations. Several exploratory approaches were removed from the final notebook so that the submitted code reflects the controlled final experiment rather than every intermediate exploration.

### 4.1 Developing the morphology-aware tokenizer

The initial plan was to run Morfessor over the RuSentiment vocabulary and use the resulting segments directly as the morphology-aware token representation of each social media post text. This raised two problems. First, unsupervised Morfessor segmentation was not linguistically reliable enough for an explainability experiment. Morfessor learns recurring patterns statistically, so a recurring string can be useful to the model without necessarily being a true Russian morpheme. If poor segments were passed directly to SHAP, the project would simply replace arbitrary BPE fragments with arbitrary Morfessor fragments and the comparison would be redundant.

The first solution was therefore to make Morfessor semi-supervised. The Morfessor toolkit allows a model to be trained using both unlabelled corpus statistics and labelled segmentations. In this project, Russian word frequencies from the RuSentiment training corpus provided the unlabelled data, while Russian SIGMORPHON ‘gold boundary’ dataset mentioned in section…. 3 was used as labelled examples of where morphological boundaries should occur. In simpler terms, RuSentiment tells Morfessor which words and recurring patterns are common in the target social-media domain, while SIGMORPHON gives the model examples of what a linguistically plausible Russian segmentation looks like. The Morfessor model is therefore trained specifically during this project using these two sources.

RuSentiment word frequencies + SIGMORPHON annotations → semi-supervised Morfessor

Before the SIGMORPHON annotations were used for training, some data cleaning was required. SIGMORPHON contains canonical morphological analyses, and because Russian inflection and stem alternation can change the written form of a word, the annotated segments do not always concatenate exactly back to the observed surface form. These analyses may still be linguistically valid, but they are unsuitable as direct supervision for a tokenizer that must place boundaries between the characters actually present in the text. It was therefore decided tha only annotations whose segments reconstructed the original word exactly were retained as training data for Morfesser.

$$m_1 + m_2 + \ldots + m_k = w \quad \rightarrow \quad \text{keep annotation}$$

Where mi is the morphemes and w is a word.

Morfessor was first considered as the sole morphology-aware token representation. This produced useful linguistic segmentations, but a classifier still requires a fixed vocabulary of token IDs. A Morfessor segment encountered during validation or testing may not have been observed during training and therefore may not have a corresponding token ID. The final approach therefore keeps every Cyrillic Morfessor segment observed in the RuSentiment training split as a whole token. If Morfessor later predicts a segment that is not in this training vocabulary, BPE is used only for that segment as a fallback. This allows known morphological units to remain intact for later SHAP analysis while preserving coverage of rare or previously unseen forms.

The algorithm below described the final morphology-aware segmentation approach which was much more complex then what the initial methodology proposed.

**Algorithm 1: Morfessor tokenizer with BPE fallback**

**Input:** normalised RuSentiment training corpus *D*, trained Morfessor model *M*
**for** each Russian word *w* in *D* do
    segment *w* using Morfessor *M*
    add each resulting Cyrillic segment to morphology vocabulary *Vm*
**end for**

train a BPE fallback vocabulary

**for** each Morfessor segment *m* during tokenisation
  **if** *m* m is a member of *Vm*
      keep *m* as one token
  **else**
      tokenise *m* using BPE
  **end if**
**end for**

**Output:** morphology-aware token sequence

Algorithm 1 summarises the final morphology-aware tokenizer. Morfessor defines the preferred linguistic units, and every Cyrillic Morfessor segment observed in the training corpus receives a direct vocabulary entry. These known segments are therefore kept whole rather than being subdivided by BPE. BPE is only used as a fallback for segments that were not observed during training, preventing unseen or noisy forms from being mapped directly to an unknown tokens. The aim is therefore to preserve morphological units where evidence exists while maintaining practical coverage for new text to then be able to analyse SHAP outputs. To keep the classifier comparison controlled, the standard BPE tokenizer is trained to the same final vocabulary size as the morphology-aware tokenizer.

## 5. Result & Discussion

### 5.1 Morphological segmentation and tokenizer quality

Before comparing sentiment explanations, the morphology-aware pipeline was evaluated to confirm that it produced linguistically plausible morphological units without creating a serious coverage problem. The semi-supervised Morfessor model achieved a held-out test boundary F1 of 0.874 and exact-match accuracy of 0.852. Recall was high at 0.966, while precision was lower at 0.797, showing that the main weakness was over-segmentation rather than missed boundaries. This was reflected in the error breakdown: 7,267 test words were segmented correctly, 964 were over-segmented, 202 had different boundaries and only 93 were under-segmented.

The final tokenizer maintained every Cyrillic Morfessor segment observed in the RuSentiment training split as a direct token and used SentencePiece BPE only as fallback for unseen segments. This produced 18,105 morphological units that were seen in training and a final vocabulary of 20,626 tokens. Standard BPE was trained with the same vocabulary size to control vocabulary capacity.

| Measure | Standard BPE | Morphology-aware |
|---|---|---|
| Test mean sequence length | 21.735 | 23.367 |
| Test sequences >128 tokens | 1.793% | 2.098% |
| Test UNK rate | 0.102% | 0.095% |
| Test BPE fallback rate | – | 0.462% |
| Morfessor boundary F1 | – | 0.874 |
| Morfessor exact match | – | 0.852 |

*Table 1. Morphological segmentation and tokenizer characteristics.*

As seen in table 1, the morphology-aware representation produced only slightly longer sequences while maintaining very high coverage. Fewer than **0.5%** of test-set Cyrillic segments required BPE fallback. Qualitative checks also showed the intended behaviour.

## 5.2 Sentiment classification performance

The two tokenizers were then evaluated through matched lightweight RoBERTa classifiers. Both models used the same architecture, training procedure, data split and final vocabulary capacity, leaving tokenisation as the main experimental difference.

| Split | Tokenizer | Accuracy | Macro F1 |
|---|---|---|---|
| Validation | Standard BPE | 0.7242 | 0.7100 |
| Validation | Morfessor + BPE fallback | 0.7336 | 0.7204 |
| Test | Standard BPE | 0.7741 | 0.7353 |
| Test | Morfessor + BPE fallback | 0.7669 | 0.7243 |

*Table 2. Sentiment classification results.*

Overall, the two classifiers achieved performed similarly. The morphology-aware model performed slightly better on validation data, while standard BPE performed slightly better in testing, where macro F1 differed by only 0.011. This small difference suggests that morphology-aware tokenisation maintained broadly comparable classification performance, allowing the main comparison to focus on explanation quality.

## 5.3 SHAP explanation analysis

The main research question asks about explanation quality rather than classification accuracy. SHAP was therefore applied to token positions, with only Cyrillic-containing token features varied during masking. Punctuation, emoji and other non-Cyrillic material remained visible to the classifier but were held fixed. The analysis does not treat larger SHAP values as automatically better; the central question is whether highly attributed features correspond to more recognisable and sentiment-relevant Russian units.

Twelve qualitative SHAP examples were inspected during the analysis. An initial set of six correctly classified validation examples was examined, followed by six additional examples selected to provide clearer tokenisation contrasts between the two conditions. The purpose was exploratory rather than to estimate the frequency of

improvement across the dataset. Four representative cases are discussed below, with the full set of qualitative outputs reported in Appendix C

Case 1: clear improvement

**“я не хочу отношении, они заебали. когда ты любишь а тебя наебали...”**

*“I don’t want relationships, I’m fed up with them. When you love someone and they screw you over...”*

Standard BPE segments наебали as нае | бали, creating the fragment бали, which has no independent meaning in this context. The morphology-aware tokenizer instead produces наебал | и, preserving the meaningful verb stem наебал-, from наебать (“to screw over/deceive”), while separating the plural ending -и. SHAP attribution therefore falls on a unit that a Russian speaker can connect directly to the negative meaning of the sentence, making the explanation more linguistically interpretable.

Case 2: better morphology, but little explanatory change

**“девушка может плакать по-разному. навзрыд, истерично, всхлипывая, ругаясь. но самый пиздец - это тихие слезы с каменным лицом и молчание. ”**

*“A girl can cry in different ways: sobbing, hysterically, sniffling, swearing. But the worst is quiet tears with a blank face and silence.”*

The morphology-aware tokenizer keeps истерично (“hysterically”) as a complete word, whereas standard BPE splits it as истери | чно. Keeping the full form makes the token easier to interpret because a Russian speaker can immediately connect it to the manner of crying described in the sentence. However, the strongest SHAP feature in both models is already the complete negative swear word пиздец, so the cleaner segmentation of истерично has only a limited effect on the overall explanation.

Case 3: no real benefit

**“сукаааа, где мой комп!!!!”**

*“F****, where is my computer!!!!”*

The main sentiment-bearing lexical item in this sentence is the swear word сука. Both tokenizers preserve сука as a complete word. In this context, сука is being used as a strong word expressing frustration, and it is also the strongest SHAP feature in both models. Because standard BPE already keeps the main sentiment-bearing word intact, the morphology-aware tokenizer does not make the explanation substantially easier to interpret.

Case 4: morphology can make things worse

**“Лёжа в ванной хорошо думать о политике =)”**

*“Lying in the bath is a good time to think about politics =)”*

In this case, standard BPE keeps ванной (“in the bath”) as one complete word, whereas Morfessor splits it as в | анной. This is misleading because в is itself a genuine Russian preposition meaning “in”, but here it is not functioning as a separate word or meaningful morpheme inside ванной. The split therefore creates an artificial feature that appears linguistically meaningful even though it is only a segmentation error, making the SHAP explanation less coherent.

Across the 12 qualitative examples inspected (see Appendix C), morphology-aware tokenisation sometimes made the evidence highlighted by SHAP easier to relate to the predicted sentiment by replacing arbitrary fragments with recognisable Russian units. However, the effect was inconsistent. This is because standardBPE

often already preserved meaningful sentiment-bearing words, while Morfessor errors could reduce interpretability.

## 6. Conclusion

This project investigated whether morphology-aware tokenisation improves the linguistic interpretability of SHAP explanations for Russian sentiment classification while maintaining comparable predictive performance.

The first part of the research question concerned predictive performance. The two classifiers performed broadly similarly. Standard BPE achieved a test macro F1 of 0.7353, compared with 0.7243 for the morphology-aware model, a difference of 0.011. The morphology-aware approach therefore did not improve classification performance, but it also did not introduce a substantial performance penalty.

The second and more important part concerned interpretability. The qualitative SHAP analysis showed that morphology-aware tokenisation could make explanations more meaningful when it preserved a recognisable Russian unit that standard BPE had fragmented. However, this benefit was conditional. Where standard BPE already preserved meaningful sentiment-bearing words, morphology added little, while incorrect Morfessor segmentation could make explanations less coherent.

Overall, the results suggest that morphology-aware tokenisation is not inherently more interpretable, but it can improve token-level explanations when the segmentation is accurate and the resulting unit is relevant to the model prediction. Because the SHAP analysis was qualitative and based on a small sample, these findings should be treated as evidence of how morphology can affect explanation quality rather than as a measure of how often improvement occurs across the full dataset.

## 7. Use of AI Declaration

I declare that any AI (or GenAI) usage within the project has been recorded and noted within Appendix B or within the main body of the text itself. This includes (but i not limited to) usage of text generation methods including LLMs, text summarisation methods, or image generation methods. I understand that failing to divulge use of AI within my work counts as contract cheating and can result in a zero mark for the ECS7036p Individual Project or even failing the module.

Signature: [Alexa Garcia Nunez]

## REFERENCES


[1] R. Sennrich, B. Haddow, and A. Birch, "Neural Machine Translation of Rare Words with Subword Units," in Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), Berlin, Germany, 2016, pp. 1715–1725. https://aclanthology.org/P16-1162/

[2] M. Creutz and K. Lagus, "Unsupervised Models for Morpheme Segmentation and Morphology Learning," ACM Transactions on Speech and Language Processing, vol. 4, no. 1, article 3, 2007. https://doi.org/10.1145/1187415.1187418

[3] T. Bauwens and P. Delobelle, "BPE-knockout: Pruning Pre-existing BPE Tokenisers with Backwards-compatible Morphological Semi-supervision," in Proceedings of NAACL-HLT 2024 (Volume 1: Long Papers), Mexico City, Mexico, 2024, pp. 5810–5832. https://aclanthology.org/2024.naacl-long.324/

[4] E. Asgari, Y. El Kheir, M. SadraeiJavaheri, and A. Nazari, "MorphBPE: Morphology-Aware Tokenization for Efficient LLM Training," in Findings of the Association for Computational Linguistics: ACL 2026, San Diego, CA, USA, 2026, pp. 41610–41621. https://aclanthology.org/2026.findings-acl.2068/

[5] K. Batsuren et al., "The SIGMORPHON 2022 Shared Task on Morpheme Segmentation," in Proceedings of the 19th SIGMORPHON Workshop on Computational Research in Phonetics, Phonology, and Morphology, Seattle, WA, USA, 2022, pp. 103–116. https://aclanthology.org/2022.sigmorphon-1.11/

[6] A. Rogers, A. Romanov, A. Rumshisky, S. Volkova, M. Gronas, and A. Gribov, “RuSentiment: An Enriched Sentiment Analysis Dataset for Social Media in Russian,” in Proceedings of the 27th International Conference on Computational Linguistics, Santa Fe, NM, USA, 2018, pp. 755–763. https://aclanthology.org/C18-1064/

[7] S. M. Lundberg and S.-I. Lee, “A Unified Approach to Interpreting Model Predictions,” in Advances in Neural Information Processing Systems 30, 2017.
https://proceedings.neurips.cc/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html

[8] A. Madsen, S. Reddy, and S. Chandar, “Post-hoc Interpretability for Neural NLP: A Survey,” ACM Computing Surveys, vol. 55, no. 8, article 155, pp. 1–42, 2022. https://doi.org/10.1145/3546577

[9] M. A. Yagiz and F. Horasan, “Beyond the Token: Correcting the Tokenization Bias in XAI via Morphologically-Aligned Projection,” in Proceedings of the Second Workshop on Natural Language Processing for Turkic Languages (SIGTURK 2026), Rabat, Morocco, 2026, pp. 228–235. https://aclanthology.org/2026.sigturk-1.19/

**Appendix A: Reflection**

This project came about from my interest in the intersection between languages, NLP and explainable AI. Since my undergraduate degree was in Russian and Mandarin, using this project to apply what I have learnt throughout this course to a subject I am so passionate about felt natural. I initially expected the main challenge to be training and comparing the sentiment classifiers, but in practice the tokenizer development became the most demanding part of the project. This was useful because it changed my understanding of tokenisation from a simple preprocessing choice into a modelling decision that can affect both prediction and the way a prediction is later explained.

The first important difficulty was that an unsupervised morphology-aware tokenizer is not automatically a good linguistic tokenizer. In the case of this project, Morfessor learns statistically recurring word parts rather than Russian grammar directly. Although this can discover real stems and affixes, it can also introduce boundaries that are statistically useful but linguistically poor. I therefore then added SIGMORPHON supervision and evaluated the resulting segmenter separately before using it in the sentiment experiment. This taught me the importance of evaluating an intermediate component rather than assuming that a method is correct because it has a linguistic label.

A second challenge was the relationship between morphology and vocabulary coverage. This project experimented with various approaches until the final methodology was confirmed. Earlier implementation tried to protect a selected subset of Morfessor segments while still using standard BPE. I thought it would be a bes t of both worlds approach. While this worked technically, the selection rules of which Morfessor segments to mantain became difficult to justify because short Russian morphemes can still be linguistically important. I eventually simplified the design so that every Cyrillic Morfessor segment observed in the training corpus receives a direct token and BPE is used only as fallback for unseen segments. This was a useful lesson in experimental design and reminded me that sometimes a simpler mechanism that directly reflects the research question is often easier to defend than a more complicated mechanism with several arbitrary thresholds.

The XAI stage also changed how I interpreted the project. I originally thought that cleaner morphological tokenisation would lead quite directly to cleaner SHAP explanations. The final examples showed that this is only partly true. A word can be segmented more linguistically without becoming important to the model, while standard BPE can already represent many important words clearly. Conversely, an incorrect morphological split can make the explanation worse. This made the final conclusion more nuanced than my original expectation and reinforced the importance of analysing and looking failure cases as well rather than only examples that support the hypothesis.

I also learnt a lot about controlled experiment comparison throughout the project. The two classifiers needed the same data splits, architecture, training settings and vocabulary capacity so that the tokenizer remained the main experimental difference. The final matched vocabulary was much larger than in my first implementation because it was determined by the number of training-seen Morfessor segments. Matching the standard BPE vocabulary to this size made the baseline stronger and therefore made the final comparison more credible, even though it reduced the number of dramatic morphology-aware wins.

Morphology aware tokenization and its relation to XAI is a very interesting complex field and this project only really touches the surface. Before starting I don’t think I realised how complex and accurate segmenter would be.

If I continued the project, I would run several random seeds, explore other techniques such as Neural network segmenters. I do think this project was a little ambitious for the available time and computational resources, but

it helped me develop practical skills in tokenizer construction, experimental debugging, model evaluation and XAI. Most importantly, I learned that interpretability depends not only on the explanation algorithm itself but also on the linguistic quality of the features that the model and explanation method are given. I really enjoyed this project and thank the course organiser for giving us the freedom to tailor our project to our areas of interest.

## Appendix B: AI Prompts/Tools

Generative AI was used as a development tool during this project. My Generative AI use mainly consisted of support with implementing the tokenizer and SHAP codes as it was new to me and also with discussing potential implementation strategies when I was transitioning from unsupervised Morfessor to semi-supervised.

| Tool / use | How it was used |
|---|---|
| ChatGPT -5.6 Sol | Help with understanding unfamiliar Morfessor and SentencePiece APIs and drafting and debugging tokenizer code. |
| ChatGPT -5.6 Sol | Discussion of experimental alternatives, including the transition from unsupervised Morfessor to semi-supervised Morfessor and the later BPE-fallback design. |
| ChatGPT -5.6 Sol | Help drafting and debugging the token-masking code used for the SHAP analysis. |
| ChatGPT -5.6 Sol | Review of executed notebook outputs to check that reported tokenizer, classifier and SHAP results matched the final implementation. |

*Table B1. Declared use of generative AI tools.*

Any AI-generated code was not treated as automatically correct or simply copies. Instead, it was used as a backbone for the algorithms I implemented and used to help me make sure I understood all the code.

## Appendix C: Full qualitative SHAP results comparison

The following 12 cases were inspected during the qualitative SHAP analysis. For each example, the Russian post is followed by an English translation or an overview for longer bits of text. There is then a table containing the comparisons of the five highest-ranked SHAP features under each tokenizer. The Russian text and SHAP output values have been taken from the executed implementation notebook.

### Case 1 (negative)

сукаааа, где мой комп!!!!

*"F***, where is my computer!!!!"*

| Rank | Standard BPE | Morfessor + BPE fallback |
|---|---|---|
| 1 | сука (+0.738) | сука (+0.772) |
| 2 | комп (+0.069) | где (+0.049) |
| 3 | где (+0.055) | а (+0.016) |
| 4 | мой (-0.022) | а (+0.016) |
| 5 | ааа (-0.019) | а (+0.016) |

*Table C1. Top five SHAP features for Case 1.*

### Case 2 (negative)

Жалко,что под стрыкало не делают дабстеп ремиксы((

*"It's a shame they don't make dubstep remixes for Strykalo ((" *

| Rank | Standard BPE | Morfessor + BPE fallback |
|---|---|---|

| | | |
|---|---|---|
| 1 | ст (-0.168) | жалко (+0.060) |
| 2 | жалко (+0.105) | делают (-0.041) |
| 3 | ик (-0.034) | дабстеп (-0.025) |
| 4 | делают (-0.028) | стр (+0.020) |
| 5 | ло (+0.025) | ы (-0.019) |

*Table C2. Top five SHAP features for Case 2.*

### Case 3 (negative)

я не хочу отношении,они заебали. когда ты любишь а тебя наебали...

*"I don't want relationships, I'm fed up with them. When you love someone and they screw you over..."*

| Rank | Standard BPE | Morfessor + BPE fallback |
|---|---|---|
| 1 | заебали (+0.335) | заебали (+0.382) |
| 2 | хочу (-0.273) | хочу (-0.276) |
| 3 | бали (+0.167) | не (+0.133) |
| 4 | любишь (-0.166) | любишь (-0.118) |
| 5 | не (+0.098) | наебал (+0.110) |

*Table C3. Top five SHAP features for Case 3.*

### Case 4 (positive)

Лёжа в ванной хорошо думать о политике =)

*"Lying in the bath is a good time to think about politics =)"*

| Rank | Standard BPE | Morfessor + BPE fallback |
|---|---|---|
| 1 | хорошо (+0.569) | хорошо (+0.607) |
| 2 | лёжа (-0.181) | лёжа (-0.157) |
| 3 | в (-0.146) | в (-0.135) |
| 4 | думать (-0.111) | в (-0.131) |
| 5 | политике (-0.072) | о (-0.094) |

*Table C4. Top five SHAP features for Case 4.*

### Case 5 (positive)

А ты меня не перестаёшь удивлять..аахахаахах))))

*"And you never stop surprising me... ahahaha))))"*

| Rank | Standard BPE | Morfessor + BPE fallback |
|---|---|---|
| 1 | хаха (+0.296) | ять (-0.170) |
| 2 | аа (+0.243) | ах (+0.133) |
| 3 | ть (-0.165) | ах (+0.120) |
| 4 | меня (+0.120) | ах (+0.114) |
| 5 | не (-0.106) | не (-0.095) |

*Table C5. Top five SHAP features for Case 5.*

### Case 6 (positive)

песня с этого фильма,очень класный фильм,рекомендую посмотреть))))

*"A song from this film; it's a really great film, I recommend watching it))))"*

| Rank | Standard BPE | Morfessor + BPE fallback |
|---|---|---|
| 1 | песня (+0.351) | песня (+0.393) |
| 2 | с (-0.340) | с (-0.325) |
| 3 | очень (+0.089) | клас (+0.147) |
| 4 | посмотреть (-0.044) | очень (+0.088) |
| 5 | рекомендую (+0.044) | посмотреть (-0.071) |

*Table C6. Top five SHAP features for Case 6.*

### Case 7 (negative)

девушка может плакать по-разному. навзрыд, истерично, всхлипывая, ругаясь. но самый пиздец - это тихие слезы с каменным лицом и молчание.

*"A girl can cry in different ways: sobbing, hysterically, sniffling, swearing. But the worst is quiet tears with a blank face and silence."*

| Rank | Standard BPE | Morfessor + BPE fallback |
|---|---|---|
| 1 | пиздец (+0.189) | пиздец (+0.426) |
| 2 | слезы (+0.050) | слезы (+0.051) |
| 3 | это (+0.031) | плакать (+0.050) |
| 4 | истери (+0.029) | лицом (+0.040) |
| 5 | плакать (+0.025) | ному (+0.030) |

*Table C7. Top five SHAP features for Case 7.*

### Case 8 (positive)

Народ! мы лучшие =) как бы пошло не звучало, но будучи сперматозоидом, мы заняли первое место и появились на свет =) а сколько было соперников...))

*"People! We are the best =) As crude as it may sound, even as sperm cells we came first and appeared in the world =) and there were so many competitors...))"*

| Rank | Standard BPE | Morfessor + BPE fallback |
|---|---|---|
| 1 | народ (-0.252) | народ (-0.078) |
| 2 | лучшие (+0.132) | как (-0.046) |
| 3 | но (+0.062) | лучшие (+0.043) |
| 4 | заняли (+0.050) | мы (+0.033) |
| 5 | сколько (-0.048) | на (-0.024) |

*Table C8. Top five SHAP features for Case 8.*

### Case 9 (positive)

Я люблю фармацевтов...о как сильно!!!!!

*“I love pharmacists... oh, so much!!!!!”*

| Rank | Standard BPE | Morfessor + BPE fallback |
|---|---|---|
| 1 | люблю (+0.374) | люблю (+0.407) |
| 2 | сильно (+0.115) | сильно (+0.173) |
| 3 | рма (-0.073) | ов (-0.092) |
| 4 | фа (+0.044) | о (-0.085) |
| 5 | тов (-0.041) | т (-0.081) |

*Table C9. Top five SHAP features for Case 9.*

### Case 10 (negative)

А он заставлял меня ревновать...каждый день...обманывал часто...я психовала...переживала,мы ругались...всегда находился повод...он часто обо мне забывал,прикрываясь постоянной суматохой..уежал на долго...я скучала..звонила,писала..и снова наталкивалась на какое-то его безразличие..потом я начала привыкать...он продолжал делать мне больно...сам не понимая,что с каждым днем я к нему холоднела...он все переводил в шутки...я же понимала..мне стало слишком похуй...он извинялся....я прощала...а вскоре я ушла...и "НАС" не стало...† делать мне больно...сам не понимая,что с каждым днем я к нему холоднела...он все пере

*Long negative post about jealousy, repeated arguments, being ignored and hurt, eventually becoming indifferent and leaving the relationship.*

| Rank | Standard BPE | Morfessor + BPE fallback |
|---|---|---|
| 1 | нему (-0.108) | ставлял (-0.108) |
| 2 | больно (+0.055) | больно (+0.067) |
| 3 | ре (-0.050) | а (-0.022) |
| 4 | скучала (+0.048) | ходил (+0.020) |
| 5 | уе (+0.042) | с (-0.017) |

*Table C10. Top five SHAP features for Case 10.*

### Case 11 (negative)

А я не знаю даже сколько в моей душе ран и трещин,разбито сердце на осколки увы любить больше нечем.....(((((

*“I don’t even know how many wounds and cracks there are in my soul; my heart is broken into pieces, and there is nothing left with which to love.....(((((”*

| Rank | Standard BPE | Morfessor + BPE fallback |
|---|---|---|
| 1 | лки (-0.006) | трещин (-0.005) |
| 2 | знаю (-0.004) | оскол (+0.004) |
| 3 | разби (+0.003) | знаю (-0.003) |
| 4 | увы (+0.002) | на (-0.003) |
| 5 | не (-0.002) | не (-0.002) |

*Table C11. Top five SHAP features for Case 11.*

### Case 12 (positive)

Все-таки старинная музыка находит удивительный отклик в глубине души. Когда звуки барабанов и волынки заставляют откликнуться внутренние струны. Мелодия как река уносит в неведомые дали фантазии. И не хочется возвращаться в реальность.

Вчера в удивительно теплой обстановке клуба MOD состоялся концерт двух прекрасных групп: Triquetrum из Питера и Очелья Сороки из Москвы. Было здорово. Замечательные и талантливые ребята. Завораживающая старинная музыка, песни, потрясающе красивый голос Ульяны и зажигательные танцы. Особенно зажигательно танцевали ребята из

морской дружины Рус во главе с Сергеем Кашиным.
Я прекрасно провела вечер пятницы.
*Long positive post describing old music, talented performers and energetic dancing, ending with a wonderful Friday evening.*

| Rank | Standard BPE | Morfessor + BPE fallback |
|---|---|---|
| 1 | здорово (+0.132) | танцы (+0.114) |
| 2 | заворажи (+0.102) | здорово (+0.113) |
| 3 | прекрасных (+0.072) | прекрасных (+0.069) |
| 4 | красивый (+0.067) | и (+0.066) |
| 5 | танцы (+0.064) | заставляют (+0.060) |

*Table C12. Top five SHAP features for Case 12.*